

Weak solution of the merged mathematical equations of the polluted atmosphere

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Abstract

Considered as a geophysical fluid, the polluted atmosphere shares the shallow domain characteristics with other natural large-scale fluids such as seas and oceans. This means that its domain is excessively greater horizontally than in the vertical dimension, leading to the classic hydrostatic approximation of the Navier-Stokes equations. The authors of the [6] article have proved a convergence theorem for this model with respect to the ocean, without considering pollution effects. In this present work we provide a generalization of their result translated to the atmosphere, extending the fluid velocity equations with an additional convection-diffusion equation representing pollutants in the atmosphere.

Keywords: Navier-Stokes equations, shallow domains, pollution evolution equation, hydrostatic approximation, compactness, weak solutions.

Contents

1	Introduction	4
1.1	Notations	5
2	The polluted atmosphere model	6
3	Weak formulation and Main Result	12
4	Proof of the main theorem	14
4.1	Energy inequality	15
4.2	A priori estimates and weak convergence	16
4.3	Passing to the limit	17
5	Convection dominated coastal model	20
6	Concluding remarks	23
A	The existence of weak solution of the hydrostatic system of the polluted atmosphere	25

1 Introduction

A widely used concept in the field of large-scale geophysical fluids is to take advantage of their approximately flat structure when modeling them with the Navier-Stokes equations. The main idea is to exploit the fact that when we are working on domains such as the ocean or the atmosphere, we are working on an "almost" two-dimensional set, and thus we can significantly simplify the model in the vertical direction, or even eliminate the third dimension and arrive to a limit model in two dimensions. A common practice for example is to depth-integrate the Navier-Stokes equations for big lakes and seas and obtain the shallow water equations. Our approach is different, we keep the vertical velocity and will arrive to a three dimensional limit model. We use the

$$\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}}$$

aspect ratio and the fact that, as the above described general ideas suggest, it is natural to take the limit model as ϵ goes to zero.

In this work we provide a sort of expansion of the convergence result of [6]. The main theorem of their article is for the case of the Navier-Stokes equations of the ocean, without the inclusion of salinity. Their primary focus is on the assumption that the pressure is hydrostatic, i.e., increases linearly in the vertical direction, and the authors justify this assumption by asymptotic analysis, taking the small parameter ϵ to zero. In the derivation of the primitive equations, one of the fundamental tools is the anisotropic viscosity hypothesis, i.e., that the ratio between the horizontal and vertical scales leads to horizontal and vertical viscosities of significantly different magnitude, specifically we can use $\nu = (\nu_1, \nu_2, \epsilon^2 \nu_3)$. With this they verify that the weak solutions of the Navier-Stokes equations converge to a weak solution of the limit problem with hydrostatic pressure.

We will use the identical anisotropic viscosities describing the atmosphere by means of the Navier-Stokes eqations, and adapt their main result for the atmospheric case, therefore their article is the main pillar and reference for this work. In this paper we are providing a generalization of their result in the sense that we obtain a theorem describing the atmospheric flow combined with pollution effects, which constitutes the main novelty part, and the merged physical phenomena can naturally be seen as the air-analogous version of the salty sea (a homogeneous fluid carrying some other particles of a certain concentration).

Thus, we investigate the convergence of the weak solutions of the pol-

luted atmosphere modeled by the incompressible anisotropic Navier-Stokes equations combined with a convection-diffusion equation, and show that in this extended case as well, the weak solutions converge to the solution of the hydrostatic limit model.

This paper is organized in the following way. In section 2 we describe the physical model of the polluted atmosphere and the scaling leading to the hydrostatic equations. In section 3 we introduce the function spaces we work in, and we present the weak formulation and the main theorem. The main theorem's proof is described in section 4, and a generalized model is introduced in chapter 5. A few concluding remarks can be found in section 6, finally in the appendix of the article we present an existence result for the hydrostatic equations of the polluted atmosphere.

We end up this section with some notations we will use through the paper.

1.1 Notations

- a) ϵ = the aspect ratio
- b) ν = viscosity
- c) Ω_ϵ, Ω = the original and the rescaled (ϵ -independent) domain, respectively
- d) $\nabla = (\partial_x, \partial_y, \partial_z)$ the gradient vector
- e) $\nabla_\nu = (\nu_x^{1/2}\partial_x, \nu_y^{1/2}\partial_y, \nu_z^{1/2}\partial_z)$
- f) $\Delta_\nu = \nu_x\partial_{xx}^2 + \nu_y\partial_{yy}^2 + \nu_z\partial_{zz}^2$ the anisotropic Laplacian
- g) $\mathbf{g}$ = represents the force due to gravity, which also includes the centrifugal effect
- h) φ = the gravity potential term, i.e. $\mathbf{g} = \nabla\varphi$
- i) $\mathbf{w}$ represents the earth rotation angular speed, $l(y)$ = latitude
- j) $\alpha = 2f \sin(l(x_2)), \beta = 2f \cos(l(x_2))$
- k) $(\cdot, \cdot)$ = the scalar product in $L^2(\Omega)^d$ or the duality $L^p(\Omega), L^{p'}(\Omega)$
- l) $\langle \cdot, \cdot \rangle_\Gamma$ = the scalar product or duality on the boundary curve Γ ($H^{-1/2}(\Gamma), H^{1/2}(\Gamma)$)

m) $\mathbf{u}_H$ = the horizontal velocity components (u_1, u_2)

n) $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$

2 The polluted atmosphere model

We begin by choosing a coordinate system, set the domain we start with and an appropriate set of equations describing both atmosphere motions and the effect of pollution. We ignore large scale effects such as the curvature of the Earth. For describing the effect of pollution we use a continuity equation written in Cartesian coordinates for the concentration.

We fix the local Cartesian coordinate system to be adjusted to the downwind direction, i.e. z is the vertical direction, while x is the downwind, and y is the crosswind direction. It is a legitimate assumption if we are considering a time interval which is not too long in the sense that it is reasonable to assume that we have a permanent, relatively stable main wind direction.

The domain we consider is a local slice of the atmosphere filled with homogeneous air that we assume to be incompressible, and a pollutant of concentration C , therefore it is defined by the set

$$\Omega_\epsilon = \{(x, y, z) \in \mathbb{R}^3; (x, y) \in \omega, 0 < z < \epsilon h(x, y)\}, \quad (1)$$

where ω is a Lipschitz-domain in $\mathbb{R}^2$, h is a nonnegative Lipschitzian application, and ϵ incorporates the shallowness of the domain.

The equations governing the air velocity $\mathbf{v}$ and pressure q in the atmosphere are the incompressible Navier-Stokes equations, in which we use different viscosities according to vertical or horizontal directions, and the density is taken to be identically equal to one:

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0, T) \quad (2)$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0, T) \quad (3)$$

Contaminants emitted into the atmosphere are transported by the wind, mixed and dispersed by turbulent behavior, and deposited onto the ground by gravity. The dispersion of pollutants is commonly modeled by a diffusive equation of the form

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2} + S, \quad (4)$$

where C is the pollution concentration.

In this equation S represents the sources and sinks of the pollutant in the atmosphere i.e. its emission, removal from the atmosphere by dry and wet deposition, and chemical reactions. We will use the approach suggested by [19] and use what is called a parametrized nascent delta function (a Gaussian impulse, specifically) to describe the source term S .

Let us consider a gas plume with intensity I originating at emission coordinates $\mathbf{x}_s$ beginning at time t_s . At the beginning we assume zero concentration. An abrupt gas emission is modeled by a source term represented as a product of an approximate delta function $\delta_\kappa(\mathbf{x} - \mathbf{x}_s)$ (centered at $\mathbf{x}_s$ with a constant parameter κ) and a Heaviside function s (switching from 0 to I at time t_s .) In fact, this source term models the continuous emission of the gas at constant rate and fixed location. Basically $s(t; t_s)$ is a function which activates the source, namely $s(t; t_s) := IH(t - t_s)$ where $H(\cdot)$ is the Heaviside step function and $0 \leq t_s \leq T$. We arrive to the complete source term model of the form

$$S_\kappa(t, t_s, x, x_s) = s(t; t_s)\delta_\kappa(\mathbf{x} - \mathbf{x}_s), \quad (5)$$

while in the limit model the source term S naturally stands for the δ distribution.

Now, we perform a vertical scaling to make the domain independent of ϵ .

$$x = x_1, \quad y = x_2, \quad z = \epsilon x_3$$

So we have the following new, ϵ - independent domain

$$\Omega = \{(x_1, x_2, x_3) \in \mathbb{R}^3; (x_1, x_2) \in \omega, 0 < x_3 < h(x_1, x_2)\}.$$

The corresponding scaling regarding the kinematic and viscosity terms are

$$\begin{aligned} v_1 &= u_1^\epsilon, & v_2 &= u_2^\epsilon, & v_3 &= \epsilon u_3^\epsilon, & p &= p^\epsilon, \\ \nu_x &= \nu_1, & \nu_y &= \nu_2, & \nu_z &= \epsilon^2 \nu_3, \end{aligned}$$

where the later embodies the different eddy viscosity hypothesis that we mentioned earlier, and p is the pressure term including the gravity potential φ , i.e., $p = q - \varphi$. Basically it can be heuristically deduced from the viscosity dimensions: typical length²/typical time, more details on the subject can be found in [7], [6].

Finally, we use an analogous rescaling methodology for the diffusivity constants in (4),

$$K_y = K_2, K_z = \epsilon^2 K_3.$$

As a second step of the rescaling process, we also exploit the fact that we chose the coordinate system in a way so that the x axis matches the downwind direction. In this case we use an additional scaling equation, which is different in nature from the previous ones in the sense that it is not derived from the shallowness of the domain but rather suggested by the fact that in the downwind direction it is natural to assume that the diffusion is negligible compared to downwind transport, i.e.

$$|u_1 \partial_{x_1} C| \gg \left| K_x \frac{\partial^2 C}{\partial x_1^2} \right| \quad (6)$$

which leads us to using

$$K_x = \epsilon K_1.$$

Finally, as we mentioned before, we use a Gaussian approximated delta function to represent the source, so it is convenient to reformulate (5) in terms of ϵ , which is the standard deviation parameter of the distribution describing the source — since later we will take the limit as ϵ goes to zero, it will result in having the δ distribution representing the hydrostatic source term, which is a physically natural choice. In other words, we choose κ to be ϵ and the following function will give shape to the source term:

$$\delta_\epsilon(\mathbf{x} - \mathbf{x}_s) = \gamma \epsilon^{-1} e^{-\frac{(\mathbf{x} - \mathbf{x}_s)^2}{\epsilon^2}};$$

and finally

$$S_\epsilon = IH(t - t_s) \gamma \epsilon^{-1} e^{-\frac{(\mathbf{x} - \mathbf{x}_s)^2}{\epsilon^2}}, \quad (7)$$

where γ is a normalizing constant.

Putting this condition and the previously described rescaling terms together to the original equations, we arrive to the merged anisotropic equations of the polluted atmosphere above inland surface:

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0 \text{ in } \Omega \times (0, T) \quad (8)$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0 \text{ in } \Omega \times (0, T) \quad (9)$$

$$\epsilon^2 \{ \partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon \} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0 \text{ in } \Omega \times (0, T) \quad (10)$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0 \text{ in } \Omega \times (0, T) \quad (11)$$

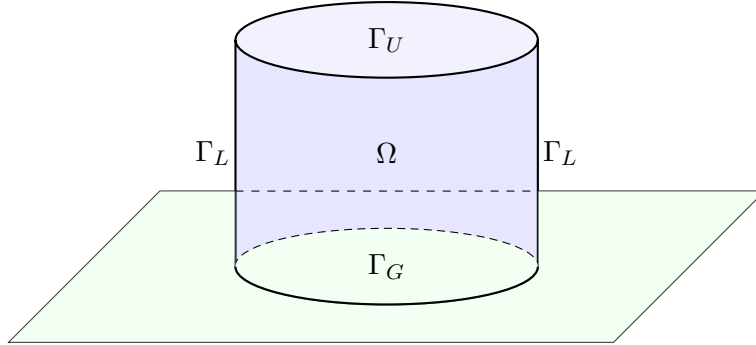
$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon \text{ in } \Omega \times (0, T) \quad (12)$$

The initial condition for the velocity and the pollution concentration are

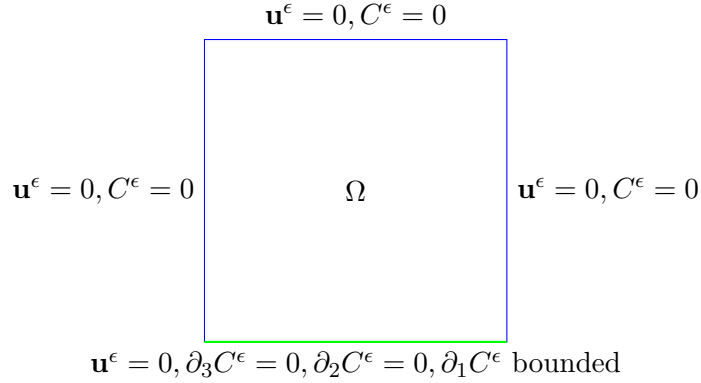
$$\mathbf{u}^\epsilon(\cdot, t=0) = \mathbf{u}_0, C^\epsilon(\cdot, t=0) = C_0 \text{ in } \Omega. \quad (13)$$

To make the above set of equations complete, we need to assign the boundary conditions.

Concerning the pollution concentration it is vital to choose boundary conditions that are physically as accurate as possible and at the same time not unnecessarily complicated. As we can see in many related papers, in the case of LAMs (limited area models) finding the right boundary conditions can be a nontrivial task.



In order to describe the boundary condition in a precise way we need to divide $\Gamma = \partial\Omega$ as follows. The upper boundary of the Ω domain is denoted by Γ_U , the lateral boundary is Γ_L , and the lower boundary of the domain is Γ_G . As we mentioned before, the case we are investigating is that of inland domains Ω , i.e. the case of landscapes and cities with no significant nearby water surface. This is very important since the boundary conditions for both wind velocity and pollution concentration in the air are different depending whether we are above solid ground or water surface.



The boundary conditions we choose to use in the inland scenario are the following:

- The upper boundary Γ_U of the domain we are considering is chosen to be at the isobar representing the planetary boundary layer ([5]), therefore we assume

$$\mathbf{u}_H = 0, u_3 = 0, C = 0 \text{ on } \Gamma_U \times (0, T) \quad (14)$$

- On Γ_L we use the same boundary conditions as on the upper level, i.e. we have

$$\mathbf{u}_H = 0, u_3 = 0, C = 0 \text{ on } \Gamma_L \times (0, T). \quad (15)$$

Remark 1. *If we wanted to use $C(x, y, z) = c$ with some c constant value, we could make a change of variables distracting that background flow and arrive back to the same mathematical problem with C being zero.*

- In this model we neglect topological variations, and the $z = 0$ plane represents the lower boundary. On the ground level we have

$$u_H = 0, u_3 = 0. \quad (16)$$

This is the mathematical way of expressing no-slip boundary conditions for the horizontal wind speed and the impervious nature of the ground with respect to the wind. As for the concentration, the way we

can choose ground level boundary conditions is the following. Firstly, we assume that the pollution is not absorbed by the ground, so we use

$$\partial_3 C = 0 \text{ on } \Gamma_G$$

(see also for example [18]). For the horizontal partial derivatives of the pollution, we can separate downwind and crosswind direction boundary conditions. In the crosswind direction we use

$$\partial_2 C = 0$$

while in the downwind (stronger) direction our assumption is more relaxed, we only assume that $\partial_1 C$ is finite in the boundary norm, i.e. in $L^2 L^2(0, T; \Gamma_G)$.

Finally, the boundary condition to describe the anisotropic system can be summarized as follows:

$$\begin{aligned} \mathbf{u}^\epsilon &= 0 \text{ on } \Gamma \times (0, T), \\ C^\epsilon &= 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \\ \partial_2 C^\epsilon &= \partial_3 C^\epsilon = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C^\epsilon \in L^2 L^2(0, T; \Gamma_G) \end{aligned} \tag{17}$$

Note that we have a domain where the lower and upper boundaries represent a physically existent layer (i.e. the ground and the planetary boundary layer), but outside the lateral boundary the physical world continues without influence and without any particular external condition. On boundaries of such nature we have to apply OBCs (open boundary conditions), which is not a trivial task, see for example [17]. We need to avoid any spurious reflection or constraint which is unnatural, but at the same time we want the model to remain mathematically manageable.

If we assume $\mathbf{u}^\epsilon = O(1)$, then neglecting the ϵ^2 and ϵ in the anisotropic equations, we formally arrive to the hydrostatic Navier-Stokes equations (i.e., primitive equations) combined with pollution.

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T) \tag{18}$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T) \tag{19}$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T) \tag{20}$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T) \quad (21)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T) \quad (22)$$

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T) \quad (23)$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1, 2 \quad (24)$$

$$\begin{aligned} C &= 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \\ \partial_2 C &= \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T), \partial_1 C \in L^2 L^2(0, T; \Gamma_G) \end{aligned} \quad (25)$$

Remark 2. Here S stands for the δ limit distribution. We keep the notation S for generality, since as we will see later, instead of the approximate delta functions and the delta distribution, it is possible to use any sort of "sufficiently smooth" source term (which can be independent from ϵ in the anisotropic case as well) that is bounded in $L_t^2 L_x^2$.

3 Weak formulation and Main Result

In this section we describe the weak formulation of the merged equations in both the anisotropic and hydrostatic case we are going to use in the paper.

We need to define the following spaces:

$$C_\Gamma^\infty(\Omega) = \{\varphi \in C^\infty(\bar{\Omega}); \varphi = 0 \text{ on some neighbourhood of } \Gamma\}$$

$$H_\Gamma^k(\Omega) = \overline{C_\Gamma^\infty(\Omega)}^{H^1(\Omega)} = \{v \in H^k(\Omega); v = 0 \text{ on } \Gamma\}$$

$$\mathbf{V} = \{\mathbf{v} \in H_\Gamma^1(\Omega) \times H_\Gamma^1(\Omega) \times H_0^1(\Omega); \nabla \cdot \mathbf{v} = 0 \text{ in } \Omega\}$$

$$H(\partial_3, \Omega) = \{v \in L^2(\Omega); \partial_3 v \in L^2(\Omega)\}$$

$$H_0(\partial_3, \Omega) = \overline{C_0^\infty(\Omega)}^{H(\partial_3, \Omega)} = \{v \in H(\partial_3, \Omega); v n_3 = 0 \text{ on } \Gamma\}$$

$$\mathbf{W} = \{\mathbf{u} \in H_\Gamma^1(\Omega) \times H_\Gamma^1(\Omega) \times H_0(\partial_3, \Omega); \nabla \cdot \mathbf{u} = 0 \text{ in } \Omega\}$$

Then the weak formulation of the hydrostatic system (18) - (25) takes the following form.

Definition 1. Find $(\mathbf{u}, C)$ such that $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and $C \in L_t^\infty L_x^2$ with $\partial_2 C, \partial_3 C \in L_t^2 L_x^2$, such that

$$\begin{aligned} & \int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla) \mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H) dt \\ & + \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u} C, \nabla \tilde{C}) + K_2(\partial_2 C, \partial_2 \tilde{C}) + K_3(\partial_3 C, \partial_3 \tilde{C}) dt = \\ & = -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt \end{aligned} \tag{26}$$

for all $(\mathbf{v}, \tilde{C})$ with $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T, \mathbf{W})$, with $v_H(T) = 0$ and $\partial_3 \mathbf{v}_H \in L^\infty(0, T; L^3(\Omega)^2)$ and $\tilde{C}$ with $\tilde{C} \in L^2(0, T, H_\Gamma^3(\Omega))$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$.

Note that the term $\int_0^T (S, \tilde{C}) dt$ makes sense since in this case S denotes the delta distribution and we have

$$(S, \tilde{C}) = \int_\Omega \delta(\mathbf{x} - \mathbf{x}_s) \tilde{C}(\mathbf{x}) d\mathbf{x} = \tilde{C}(\mathbf{x}_s).$$

Remark 3. If S is chosen to be different from the limit of the Gaussian nascent delta function, then it is easy to see that the sufficient assumption we have to make on a general S hydrostatic pollution function is that it should be in $L_t^2 L_x^2$.

The existence of weak solution for the hydrostatic system (18) - (25) can be proved in the spirit of [16]. In appendix A we will outline the main steps of the proof and the differences compared to the ocean model.

The weak form of the anisotropic system is as follows.

Definition 2. Find $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0, T; \mathbf{V}) \cap L^\infty(0, T; L^2(\Omega)^3)$ and C^ϵ with $C^\epsilon \in L_t^\infty L_x^2 \cap L^2(0, T; H^1)$, such that

$$\begin{aligned}
& \int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla) \mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\
& + \epsilon \beta \left[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3) \right] dt \\
& + \epsilon^2 \left[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3) dt \right] \\
& + \int_0^T -(C^\epsilon, \partial_t \tilde{C}) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) + \\
& \epsilon K_1 (\partial_1 C^\epsilon, \partial_1 \tilde{C}) - \epsilon K_1 [(\partial_1 C^\epsilon) \tilde{C}]_{\Gamma_G} + K_2 (\partial_2 C^\epsilon, \partial_2 \tilde{C}) + K_3 (\partial_3 C^\epsilon, \partial_3 \tilde{C}) dt = \\
& = -(\mathbf{u}_{0H}^\epsilon, \mathbf{v}_H(0)) - \epsilon^2 (u_{03}^\epsilon, v_3(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C}) dt
\end{aligned} \tag{27}$$

for all $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T; \mathbf{V})$ with $\mathbf{v}(T) = 0$ and $\tilde{C}$ such that $\tilde{C} \in H^1(0, T, H_\Gamma^3(\Omega))$ and $\tilde{C}(T) = 0$.

Note that since in our case the source term S_ϵ is given by (5), thus the last term in (27) is well defined. If we use a source function of a different and more general form, it needs to satisfy the $L_t^2 L_x^2$ boundedness criteria for the weak formulation to make sense.

We will omit the proof of the existence of weak solutions (see Definition 2.) for the anisotropic system (8) - (17) since it follows by standard finite-dimensional Galerkin approximation.

Now we are ready to state our main result.

Theorem 1. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0$, $\mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C_0 \in L^2(\Omega)^3$, $C_0 = 0$ on $\partial\Omega$; then as the aspect ratio ϵ tends to zero, any weak solution $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations (8) - (17) converge to a weak solution $(\mathbf{u}, C)$ of the hydrostatic equations of the polluted atmosphere (18) - (25).*

4 Proof of the main theorem

This section is devoted to the proof of Theorem 1. In the extended case with the pollution equation the proof relies on a priori estimates gained from the energy inequality and, as we will see this is sufficient to pass to the limit in the linear terms. Concerning the nonlinear terms that do not include

pollution we will follow [6]. What we need to work on is how to handle the nonlinear terms including the new function C^ϵ . The challenge caused by these nonlinear terms will be met by proving that a compactness result from [11] is applicable to handle them.

4.1 Energy inequality

As we mentioned before, one of the most fundamental tools in this proof is the energy inequality. If we calculate it for the anisotropic system, we arrive to the following result. For a.e. $t \in [0, T]$ the following holds:

$$\begin{aligned} & \frac{1}{2} (\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2 \|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\ & \int_0^t \{\epsilon K_1 \|\partial_1 C^\epsilon\|_{L^2}^2 + K_2 \|\partial_2 C^\epsilon\|_{L^2}^2 + K_3 \|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \int_0^t (S^\epsilon, C^\epsilon) d\tau + \\ & \frac{1}{2} (\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(0)\|_{L^2}^2 + \|C_0^\epsilon\|_{L^2}^2) + \int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon \rangle_{\Gamma_G} d\tau = \\ & I_1 + I_2 + I_3. \end{aligned} \tag{28}$$

From the hypothesis on the initial data and since the S^ϵ source term is in $L_x^\infty(\Omega)$, I_1 and I_2 are bounded. The only term we need to work on further is I_3 .

By using the Holder inequality for any $\gamma > 0$, we can estimate this term in the following way.

$$\begin{aligned} I_3 &= \int_0^t \epsilon K_1 \langle \partial_1 C^\epsilon, C^\epsilon \rangle_{\Gamma_G} d\tau \leq K_1 \int_0^t \sqrt{\epsilon} \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)} \sqrt{\epsilon} \|C^\epsilon\|_{L^2(\Gamma_G)} d\tau \leq \\ & \frac{1}{\gamma} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 d\tau + \gamma K_1 \int_0^t \frac{\epsilon}{2} \|C^\epsilon\|_{L^2(\Gamma_G)}^2 d\tau \leq \\ & \frac{1}{\gamma} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 d\tau + \gamma \frac{\epsilon}{2} K_1 c_T \int_0^t \|C^\epsilon\|_{W^{1,2}(\Omega)}^2 d\tau \leq \\ & \frac{1}{\gamma} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 d\tau + \gamma \frac{\epsilon}{2} K_1 c_T (c_S + 1) \int_0^t \|\nabla C^\epsilon\|_{L^2(\Omega)}^2 d\tau, \end{aligned} \tag{29}$$

where c_T and c_S are the constants provided by the trace theorem and Sobolev embeddings, and the boundary term is bounded because of the a priori assumptions made on the boundary conditions.

The energy inequality (28) now takes the form

$$\begin{aligned}
& \frac{1}{2} (\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2 \|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
& \int_0^t \{(\epsilon K_1 - \delta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_1 C^\epsilon\|_{L^2}^2 + (K_2 - \delta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_2 C^\epsilon\|_{L^2}^2 + \\
& + (K_3 - \delta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) \|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \frac{1}{\delta} K_1 \frac{\epsilon}{2} \int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \\
& + \frac{1}{2} (\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) + \int_0^t (S^\epsilon, C^\epsilon) d\tau.
\end{aligned} \tag{30}$$

The right hand side of the energy inequality is now apparently bounded, and it is easy to see that the coefficients of the pollution terms are positive, since the δ constant of the generalized Young inequality can be freely chosen in a way such that the requirements $\epsilon K_1 (1 - \delta \frac{1}{2} c_T (c_S + 1)) > 0$, $(K_2 - \delta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) > 0$, $(K_3 - \delta \frac{\epsilon}{2} K_1 c_T (c_S + 1)) > 0$ are all satisfied.

4.2 A priori estimates and weak convergence

From the final form of the energy inequality we obtain a priori estimates that we summarize in the following proposition.

Proposition 1. *The sequences $u_1^\epsilon, u_2^\epsilon, \epsilon u_3^\epsilon$ are bounded in $L^\infty(0, T; L^2(\Omega)) \cap L^2(0, T; H^1(\Omega))$, C^ϵ is bounded in $L^\infty(0, T; L^2(\Omega))$; furthermore the sequences $\sqrt{\epsilon} \partial_1 C^\epsilon$, $\partial_2 C^\epsilon$, and $\partial_3 C^\epsilon$ are bounded in $L^2(0, T; L^2(\Omega))$.*

As a consequence, up to there are subsequences still denoted by the same way, for which we have the following convergence results.

$$\mathbf{u}_H^\epsilon \rightharpoonup \mathbf{u}_H \quad \text{weakly in } L^\infty(0, T; L^2(\Omega)) \cap L^2(0, T; H^1(\Omega)) \tag{31}$$

$$\begin{aligned}
\epsilon u_3^\epsilon & \rightharpoonup f \quad \text{weakly for some limit element } f \\
& \text{in } L^\infty(0, T; L^2(\Omega)) \cap L^2(0, T; H^1(\Omega))
\end{aligned} \tag{32}$$

$$C^\epsilon \rightharpoonup C \quad \text{weakly in } L_t^\infty L_x^2 \tag{33}$$

$$\partial_2 C^\epsilon \rightharpoonup \partial_2 C \quad \text{weakly in } L_t^2 L_x^2 \tag{34}$$

$$\partial_3 C^\epsilon \rightharpoonup \partial_3 C \quad \text{weakly in } L_t^2 L_x^2 \tag{35}$$

$$\sqrt{\epsilon} \partial_1 C^\epsilon \rightharpoonup e \quad \text{weakly for some limit element } e \text{ in } L_t^2 L_x^2 \tag{36}$$

4.3 Passing to the limit

Taking the limit for the terms that do not include C^ϵ had been worked out before. In the linear terms the weak convergence guaranteed by the a priori estimates will be sufficient. In the nonlinear term we need an additional result that provides a weak convergence of the product of functions for which we only have weak convergence separately.

For the linear terms we can directly apply the weak convergence results of the previous section.

- (i) The convergence of the $(C^\epsilon, \frac{\partial \tilde{C}}{\partial t})$ term is easy to see, since we have $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$.
- (ii) For the next term we have $\epsilon K_1 \int_0^T (\partial_1 C^\epsilon, \partial_1 \tilde{C}) = \sqrt{\epsilon} K_1 \int_0^T (\sqrt{\epsilon} \partial_1 C^\epsilon, \partial_1 \tilde{C})$, which goes to 0 as $\epsilon \rightarrow 0$ because of the Cauchy-Schwartz inequality, since we have $\sqrt{\epsilon} \partial_1 C^\epsilon$ bounded in $L_t^2 L_x^2$, and $\tilde{C}$ is a test function.
- (iii) As we have seen before, we have $\partial_2 C^\epsilon \rightharpoonup \partial_2 C$ in $L_t^2 L_x^2$, which by definition provides that

$$\int_0^T (\partial_2 C^\epsilon, f) dt \rightarrow \int_0^T (\partial_2 C, f) dt \quad \forall f \in L_t^2 L_x^2.$$

- (iv) Analogously,

$$\int_0^T (\partial_3 C^\epsilon, f) dt \rightarrow \int_0^T (\partial_3 C, f) dt \quad \forall f \in L_t^2 L_x^2.$$

- (v) The $\epsilon K_1 \int_0^T [(\partial_1 C^\epsilon) \tilde{C}]_{\Gamma_G} dt$ line integral vanishes as $\epsilon \rightarrow 0$ since by the Cauchy-Schwartz equation we can estimate the integral by two bounded terms ($\partial_1 C^\epsilon$ was assumed to be finite in norm on the lower boundary).
- (vi) To prove $(S^\epsilon, \tilde{C}) \rightarrow (S, \tilde{C})$, we need to show that

$$\lim_{\epsilon \rightarrow 0} \int_{\Omega} S^\epsilon(\mathbf{x}, \mathbf{x}_s) \cdot \tilde{C}(\mathbf{x}) d\mathbf{x} \rightarrow \tilde{C}(\mathbf{x}_s).$$

Using the substitution $\mathbf{y} = (\mathbf{x} - \mathbf{x}_s)/\epsilon$ we have

$$\int_{\Omega} \gamma \epsilon^{-1} e^{\frac{-(\mathbf{x} - \mathbf{x}_s)^2}{\epsilon^2}} \tilde{C}(\mathbf{x}) d\mathbf{x} = \int_{\Omega_{\mathbf{y}}} \gamma e^{-\mathbf{y}^2} \tilde{C}(\epsilon \mathbf{y} + \mathbf{x}_s) d\mathbf{y}$$

This term can be dominated by $\|\tilde{C}\|_\infty \cdot \int_{\Omega_{\mathbf{y}}} \gamma e^{-\mathbf{y}^2} \tilde{C} \, d\mathbf{y} = \|\tilde{C}\|_\infty$, so we can apply the dominated convergence theorem using that the test function $\tilde{C} \in L_t^2 L_x^\infty$ and arrive to

$$\lim_{\epsilon \rightarrow 0} (S^\epsilon, \tilde{C}) = \int_{\Omega_{\mathbf{y}}} \lim_{\epsilon \rightarrow 0} \gamma e^{-\mathbf{y}^2} \tilde{C}(\epsilon \mathbf{y} + \mathbf{x}_s) \, d\mathbf{y} = \int_{\Omega_{\mathbf{y}}} \gamma e^{-\mathbf{y}^2} \tilde{C}(\mathbf{x}_s) \, d\mathbf{y} = \tilde{C}(\mathbf{x}_s),$$

using the continuity of $\tilde{C}$.

As for the nonlinear term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$, we need a compactness theorem to take its limit.

Remark 4. *As a matter of fact, we actually have three terms here, namely $(u_i^\epsilon C^\epsilon, \partial_i \tilde{C})$ for $i = 1, 2, 3$, and for the values 1, 2 of i the weak convergence of the $u_i^\epsilon C^\epsilon$ product is given by the fact that, as it is proved in the reference article, we actually have strong convergence for the horizontal velocities, and because of the a priori estimate from the energy bound we know we have weak convergence for the C^ϵ term. But this argument does not solve the challenge caused by nonlinearity because it does not hold for u_3 , so we still need an additional compactness theorem as mentioned before.*

In this case the Aubin-Lions lemma can not be applied since it would require for C^ϵ to be in a compactly embedded space such as $L^2 H^1$, which is not the case here because we only know boundedness results for $\sqrt{\epsilon} \partial_1 C$, but not for $\partial_1 C$.

We will show that the following lemma from [11] is applicable in our case.

Lemma 1. *Let g_n, h_n converge weakly to g, h respectively in $L^{p_1}(0, T; L^{p_2}(\Omega))$, $L^{q_1}(0, T; L^{q_2}(\Omega))$ where $1 \leq p_1, p_2 \leq +\infty$,*

$$\frac{1}{p_1} + \frac{1}{q_1} = \frac{1}{p_2} + \frac{1}{q_2} = 1.$$

We assume in addition that

- $\frac{\partial g^n}{\partial t}$ *is bounded in $L^1(0, T; W^{-m, 1}(\Omega))$ for some $m \geq 0$ independent of n .*
- $\|h^n - h^n(\cdot + \xi, t)\|_{L^{q_1}(0, T; L^{q_2}(\Omega))} \rightarrow 0$ *as $|\xi| \rightarrow 0$, uniformly in n .*

Then, $g^n h^n$ converges to gh (in the sense of distributions on $\Omega \times (0, T)$).

What we will show in the following is that for the functions u_i^ϵ and C^ϵ the assumptions of Lemma 1 are satisfied, which means that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ holds.

Lemma 2. *We have $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$.*

Proof. We have seen from the a priori estimates that $C^\epsilon \rightharpoonup C$ in $L_t^\infty L_x^2$, and we know from the reference article that $u_i^\epsilon \rightharpoonup u_i$ in $L_t^2 L_x^2$.

Since we are on a finite domain and thus $C^\epsilon \in L^\infty L^2$ implies $C^\epsilon \in L^2 L^2$, let us choose $q_1 = q_2 = p_1 = p_2 = 2$.

• **Bound for the time derivative.**

$$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon$$

On one hand, the S^ϵ Gaussian source term is smooth, and we see that the second part of the right hand side is in $W^{-1,2}$, simply using that $\partial_1 \sqrt{\epsilon} C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$ are in L^2 . On the other hand, the first term is in $W^{-1,1}$, because we have

$$u^\epsilon \nabla C^\epsilon = \nabla \cdot (u^\epsilon C^\epsilon) - \nabla \cdot (u^\epsilon) C,$$

and we only have to use that the divergence of u^ϵ is zero and $u^\epsilon C^\epsilon$ is in L_x^1 .

Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$.

- **Equicontinuity.** We need to show that $\|u_i^n - u_i^n(\cdot + \xi, t)\|_{L^2 L^2} \rightarrow 0$ as $\xi \rightarrow 0$. Since in the time domain the condition naturally holds, we will focus on the space part of the proof.

First we show that for a function f in the function space C_0^∞ the condition holds.

$$\begin{aligned} \|f(x + \xi) - f(x)\|_{L^2} &= \left(\int_{\Omega} |f(x + \xi) - f(x)|^2 dx \right)^{1/2} \leq \\ &\left(\int_{\Omega} (\sup_x |f(x + \xi) - f(x)|)^2 dx \right)^{1/2} = \|f(x + \xi) - f(x)\|_{L^\infty} \cdot |\Omega|^{1/p}. \end{aligned}$$

So we have

$$\lim_{\xi \rightarrow 0} \|f(x + \xi) - f(x)\|_{L^2} \leq |\Omega|^{1/p} \cdot \lim_{\xi \rightarrow 0} (\sup_x |f(x + \xi) - f(x)|) = 0,$$

since $f \in C_0^\infty$.

Now we show that the condition holds for any function $f \in L^2$. Here we exploit the fact that C_0^∞ is dense in the L^p function space. By definition this means that $\exists \varphi_n \in C_0^\infty(\Omega)$ such that $\varphi_n \rightarrow f$ in $L^p(\Omega)$. With this we have

$$\begin{aligned} & \|f(x + \xi) - f(x)\|_{L^2} = \\ & \|f(x + \xi) - \varphi_n(x + \xi) + \varphi_n(x + \xi) - \varphi_n(x) + \varphi_n(x) - f(x)\|_{L^2} \leq \\ & \|f(x + \xi) - \varphi_n(x + \xi)\|_{L^2} + \|\varphi_n(x + \xi) - \varphi_n(x)\|_{L^2} + \|\varphi_n(x) - f(x)\|_{L^2}. \end{aligned}$$

The first and third term goes to 0 as $\xi \rightarrow 0$, since $\varphi_n \rightarrow f$ in $L^p(\Omega)$. On the other hand in the first part of the proof we have shown exactly that the second term goes to 0. Which proves that $\|u_i^n - u_i^n(\cdot + \xi, t)\|_{L^2 L^2} \rightarrow 0$ as $\xi \rightarrow 0$. Eventually this means that we can apply the lemma with $h^n = u_i^\epsilon, g^n = C^\epsilon$, which proves the statement of the lemma.

□

We finally arrive to

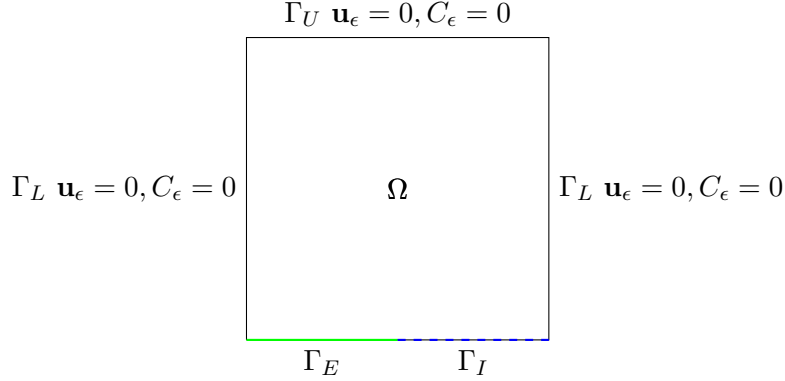
$$\int_0^T (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) dt \rightarrow \int_0^T (\mathbf{u} C, \nabla \tilde{C}) dt,$$

which, combined with the previously described linear terms, proves the main theorem.

5 Convection dominated coastal model

We now present a generalized model, which is an extension of the previous inland model by adapting the model for seaside conditions.

On the ground layer we naturally separate the boundary into two different parts in this model, $\Gamma_G = \Gamma_E \cup \Gamma_I$; Γ_E denoting the part of the boundary above the ground, and Γ_I denoting the section that is interacting with the sea. While above ground parts it is reasonable to use no-slip conditions,



above the ocean it is natural to assume that the wind has an interaction with the waves, there is no such rigidness as in the ground case. For this cause we will include θ , which represents the wind traction on the ocean surface. Analogously, we need to make similar changes for the pollution concentration as well, above sea territory it is not natural to use $\partial_3 C = 0$, since the particles can now sink under the boundary level.

On Γ_E we have the same boundary conditions as in the inland case, $\mathbf{u}^\epsilon = 0$, and for the concentration we have $\partial_3 C^\epsilon = \partial_2 C^\epsilon = 0$, and $\partial_1 C^\epsilon$ being bounded. On Γ_I on the other hand we have $\nu_3 \partial_3 \mathbf{u}_H^\epsilon = \theta_H, u_3 = 0$ with assuming a finite wind energy, i.e. the function θ being in $L^2(0, T; H^{-1/2}(\Gamma_I))$. For the pollution concentration we use the same approach and assume that the functions $\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$ are equal to some sufficiently regular, ϵ -independent functions defined on the respective boundary. More accurately, we assume that we have a $C_{\Gamma_G}^\epsilon = \sigma$ sediment function on the lower boundary, where $\sigma \in L^2(0, T; H^1(\Gamma_G))$. If the partial derivatives of σ are denoted by σ_1, σ_2 and σ_3 , we have $\partial_1 C^\epsilon = \sigma_1$ on Γ_G , and $\partial_2 C^\epsilon = \sigma_2, \partial_3 C^\epsilon = \sigma_3$ on Γ_I . (Of course σ has to be a function such that we have σ_2 and σ_3 equal to zero on the Γ_E part).

In what follows we highlight the parts that will be different from the inland case.

The weak formulation of the coastal generalized system will be different from the original inland version in that *a)* it will contain a boundary term for the wind because of the air - ocean interface, and *b)* it will contain boundary terms for $\partial_2 C, \partial_3 C$, which had been assumed to be zero before on the ground.

Calculating the weak formulation of the anisotropic system for the coastal model we arrive to the equation below.

Find $\mathbf{u}^\epsilon = (\mathbf{u}_H, u_3^\epsilon) \in L^2(0, t; \mathbf{V}) \cap L^\infty(0, T; L^2(\Omega)^3)$ and C^ϵ with $C^\epsilon \in L^\infty L^2$, $C^\epsilon \in L^2(0, T; H^1)$, $C^\epsilon(0) \in L_x^2$ such that

$$\begin{aligned}
& \int_0^T -(\mathbf{u}_H^\epsilon, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H^\epsilon, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H^\epsilon, (\mathbf{u}^\epsilon \cdot \nabla) \mathbf{v}_H) + (b(\mathbf{u}_H^\epsilon), \mathbf{v}_H) dt \\
& + \epsilon \beta \left[\int_0^T (u_3^\epsilon, v_1) - (u_1^\epsilon, v_3) \right] dt \\
& + \epsilon^2 \left[\int_0^T -(u_3^\epsilon, \partial_t v_3) + (\mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon, v_3) + (\nabla_\nu u_3^\epsilon, \nabla_\nu v_3) dt \right] \\
& + \int_0^T -(C^\epsilon, \partial_t \tilde{C}) - (\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C}) + \epsilon K_1 (\partial_1 C^\epsilon, \partial_1 \tilde{C}) - \epsilon K_1 \langle \sigma_1, \tilde{C} \rangle_{\Gamma_G} + \\
& K_2 (\partial_2 C^\epsilon, \partial_2 \tilde{C}) - K_2 \langle \sigma_2, \tilde{C} \rangle_{\Gamma_I} + K_3 (\partial_3 C^\epsilon, \partial_3 \tilde{C}) - K_3 \langle \sigma_3, \tilde{C} \rangle_{\Gamma_I} dt = \\
& = -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - \int_0^T \langle \theta_H, \mathbf{v}_H \rangle_{\Gamma_I} - \epsilon^2 (u_{03}, v_3(0)) - (C^\epsilon(0), \tilde{C}(0)) + \int_0^T (S^\epsilon, \tilde{C}) dt
\end{aligned} \tag{37}$$

for all $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T; \mathbf{V})$ with $\mathbf{v}(T) = 0$ and $\tilde{C}$ such that $\tilde{C} \in H^1(0, T, H_\Gamma^3(\Omega))$ and $\tilde{C}(T) = 0$.

On the other hand, the weak formulation of the hydrostatic case will take the following form.

Find $(\mathbf{u}, C)$ such that $\mathbf{u} = (\mathbf{u}_H, u_3) \in L^2(0, T; \mathbf{W})$ with $\mathbf{u}_H \in L^\infty(0, T; L^2(\Omega)^2)$, and C with $\partial_2 C, \partial_3 C \in L_t^2 L_x^2$, $C \in L^\infty L^2$ and $C(0) \in L_x^2$, such that

$$\begin{aligned}
& \int_0^T -(\mathbf{u}_H, \partial_t \mathbf{v}_H) + (\nabla_\nu \mathbf{u}_H, \nabla_\nu \mathbf{v}_H) - (\mathbf{u}_H, (\mathbf{u} \cdot \nabla) \mathbf{v}_H) + (b(\mathbf{u}_H), \mathbf{v}_H) dt \\
& + \int_0^T -(C, \partial_t \tilde{C}) - (\mathbf{u} C, \nabla \tilde{C}) + \\
& K_2 (\partial_2 C, \partial_2 \tilde{C}) - K_2 \langle \sigma_2, \tilde{C} \rangle_{\Gamma_I} + K_3 (\partial_3 C, \partial_3 \tilde{C}) - K_3 \langle \sigma_3, \tilde{C} \rangle_{\Gamma_I} dt = \\
& = -(\mathbf{u}_{0H}, \mathbf{v}_H(0)) - \int_0^T \langle \theta_H, \mathbf{v}_H \rangle_{\Gamma_I} - (C(0), \tilde{C}(0)) + \int_0^T (S, \tilde{C}) dt
\end{aligned} \tag{38}$$

for all $(\mathbf{v}, \tilde{C})$ with $\mathbf{v} = (\mathbf{v}_H, v_3) \in H^1(0, T, \mathbf{W})$, with $v_H(T) = 0$ and $\partial_3 \mathbf{v}_H \in L^\infty(0, T; L^3(\Omega)^2)$ and $\tilde{C}$ with $\tilde{C} \in L^2(0, T, H_\Gamma^3(\Omega))$, $\tilde{C} \in H^1(0, T, L^2)$, $\tilde{C}(T) = 0$.

Naturally, the analogous terms that we saw were different in the weak formulation will be different also in the energy inequality as well. The coastal, generalized version of the energy inequality becomes

$$\begin{aligned}
& \frac{1}{2} (\|\mathbf{u}_H^\epsilon(t)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(t)\|_{L^2}^2 + \|C^\epsilon(t)\|_{L^2}^2) + \int_0^t \{\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2 \|\nabla_\nu u_3^\epsilon\|_{L^2}^2\} d\tau + \\
& \int_0^t \{\epsilon K_1 \|\partial_1 C^\epsilon\|_{L^2}^2 + K_2 \|\partial_2 C^\epsilon\|_{L^2}^2 + K_3 \|\partial_3 C^\epsilon\|_{L^2}^2\} d\tau \leq \\
& \frac{1}{2} (\|\mathbf{u}_H^\epsilon(0)\|_{L^2}^2 + \epsilon^2 \|u_3^\epsilon(0)\|_{L^2}^2 + \|C^\epsilon(0)\|_{L^2}^2) + \int_0^t (S^\epsilon, C^\epsilon) d\tau - \int_0^t \langle \theta_H, \mathbf{u}_H \rangle_{\Gamma_I} \\
& \int_0^t \epsilon K_1 \langle \sigma_1, \sigma \rangle_{\Gamma_G} + K_2 \langle \sigma_2, \sigma \rangle_{\Gamma_I} + K_3 \langle \sigma_3, \sigma \rangle_{\Gamma_I} d\tau.
\end{aligned} \tag{39}$$

Since the σ boundary function is in $L^2 H^1$, we have boundedness on the right hand side. The inequality provides the same estimates and weak convergence results. As for passing to the limit as $\epsilon \rightarrow 0$ in the coastal version of the anisotropic system seen above, there is no mathematical novelty compared to the inland case.

Eventually we arrive to the extended, coastal version of the main theorem, which is stated below.

Theorem 2. *Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0$, $\mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; let $\theta_1, \theta_2 \in L^2(0, T; H^{-1/2}(\Gamma_I))$ and $\sigma \in L^2(0, T; H^1(\Gamma_G))$, there exists a weak solution $(\mathbf{u}, C)$ of the hydrostatic coastal equations of the polluted atmosphere (38), obtained as a limit of weak solutions $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic coastal equations (37) as the aspect ratio ϵ tends to 0.*

Remark 5. *If we switch to the "informationless", fix coordinate system that does not match the downwind direction and we drop the assumption on the flow being convection dominated (ϵ - diffusion), then we can relax the restrictions on C^ϵ on the boundary. This is true because in this case we do not have a disappearing ϵ term in front of the K_1 constant, and we can use the analogous estimate as in the inland case at the energy inequality and arrive to having a $(K_1 - \delta \frac{K_1+K_2+K_3}{2} c_T(c_S + 1)) \|\partial_1 C^\epsilon\|_{L^2}^2$ term with a positive coefficient on the left hand side and a bounded $\frac{1}{\delta_i} \frac{K_i}{2} \int_0^t \|\partial_i C^\epsilon\|_{L^2(\Gamma_G)}^2$ integral on the right hand side — the only assumption we need is $\partial_i C^\epsilon$ to be in $L_t^2 L_x^2$ on the boundary.*

6 Concluding remarks

We are finishing this work briefly mentioning a few ideas regarding the physical aspects of the model and some additional choices concerning the

source term specifically.

Remark 6. In the paper of [16] it is suggested that the domain we work on has the form of $p_0 < p < p_1$, because for the thin portion of air between the earth and the isobar $p = p_1$, another specific simplified model would be necessary. This means that a way to make the model more precise is to cut the domain at an isobar a little bit off the physical ground. This would of course radically change the boundary conditions we use for the functions (in fact it is not trivial how to change them for this case), because it would take away the advantage of having a wall-like, natural and physical boundary condition for the domain we work on. If we are positively above the ground, $\partial_3 C = 0$ would represent a strange reflective layer.

Remark 7. Note that the aspect ratio going to zero is not only legitimate on a global scale. When we switch from a global model to a local one in order to be able to use the beta plane approximation, we still have hundreds of thousands of square kilometers versus 1—5 kilometers.

Remark 8. The [16] article works with the beta plane approximation, which is appropriate for example for midlatitude regional studies, where the curvature is minimal. If we take this as the starting idea, and consider a local part of the ocean, we will most likely not have genuine physical lateral boundaries for the ocean water in the model, such as would be a coast for example. Nevertheless what the article uses as a lateral boundary condition is the salt flux being zero, which represents a wall-like boundary. What would possibly be more natural is the salt flux being constant, but it would naturally make things mathematically more complicated on the other hand. This is another example of how challenging it can be to find the right boundary terms when we are handling OBCs.

Remark 9. We chose the Gaussian nascent delta function to give shape to the source term, but it can be defined in several different ways using the many different pulses that approximate the delta function (choosing only those however that are physically meaningful in our case, for example its values are nonnegative), e.g.

- a) the unit impulse: $\delta_\epsilon(\mathbf{x} - \mathbf{x}_s) = \epsilon/2, |(\mathbf{x} - \mathbf{x}_s)| < 1/\epsilon,$
- b) the Lorentzian pulse: $\delta_\epsilon(\mathbf{x} - \mathbf{x}_s) = \epsilon/(1 + \pi^2\epsilon^2(\mathbf{x} - \mathbf{x}_s)^2).$

The Gaussian and Lorentzian version of the source term is continuous and bounded on Ω , it is actually in L^∞ in space (and in L^2 as a consequence),

while it is easy to see that the piecewise constant unit pulse is in L^2 as well. Since the time dependence of this source is described by the Heaviside function, the time norm is also finite.

A The existence of weak solution of the hydrostatic system of the polluted atmosphere

What we have showed is that the weak solutions of the anisotropic system converge to the weak solution of the hydrostatic system of the polluted atmosphere. Without having an existence theorem for the hydrostatic equations, the weak solutions of the anisotropic system could just be converging to the empty set. The reference article for unpolluted fluids states only the convergence part as well, but the existence is implicitly assumed.

To complete our work of expanding the original result to the case of a fluid containing a certain pollution concentration, we also need an existence theorem.

The idea is that the polluted atmosphere can be viewed as the air-analogous version of the salty sea water in terms of the equations that represent it. In both cases we have a fluid which can be seen approximately as a homogenous incompressible fluid carrying some other particles in a significant concentration, the basis fluid motion is well described by the incompressible Navier-Stokes equations, and the additional concentration is described by an advection-diffusion equation.

The paper of [16] provides an existence result for an arbitrary time interval $[0, t_1]$ for the system describing the salty ocean water with temperature. The structure of the equations themselves describing the system is essentially the same, if we drop the equation describing the temperature, we basically arrive to the same system that describes the polluted atmosphere.

The main difference consists in the boundary conditions: on every boundary they use a zero flux for the salinity, $\frac{\partial S}{\partial \mathbf{n}_s} = 0$. This coincides with our lower boundary conditions, but on the lateral boundary we use a fix constant.

The weak formulation we want to prove an existence theorem for (see equation (26)) is different from the equation the article provides the result for:

$$\left(\frac{d}{dt}U, \tilde{U}\right)_H + a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U}) = l(\tilde{U}),$$

where the definition of the above terms can be found on page 21 in the

article of [16]. In this version the weak formulation is only performed in space, and the space derivatives are not in every case translated to the test function when it is possible.

What the above equation means for example for the salinity is

$$\begin{aligned} & \int_{\mathcal{M}} K_S \frac{d}{dt} S \cdot \tilde{S} d\mathcal{M} + K_S \int_{\mathcal{M}} (\mu_s \nabla S \nabla \tilde{S} + \nu_s \frac{\partial S}{\partial z} \frac{\partial \tilde{S}}{\partial z}) d\mathcal{M} + \\ & + K_S \int_{\mathcal{M}} (v \nabla S + w \frac{\partial S}{\partial z}) \tilde{S} d\mathcal{M} = K_S \int_{\mathcal{M}} F_S \tilde{S} d\mathcal{M} + (K_S \mu_s \frac{\partial S}{\partial x} \tilde{S})_{\Gamma_i}. \end{aligned}$$

We will follow step by step the structure of the proof in [16], redefining the necessary quantities to be valid for our case and verify that the proof holds in the case of the polluted atmosphere as well (this is necessary not only because of using altered boundary conditions but also because the regularities of the functions are different in some cases as mentioned above).

Step 1. Firstly we redefine the norms in order to make them match our system.

$$\begin{aligned} ((U, \tilde{U})) &= ((v, \tilde{v}))_1 + K_C ((C, \tilde{C}))_2 \\ ((v, \tilde{v}))_1 &= \int_{\Omega} \nabla_{\nu} v \nabla_{\nu} \tilde{v} d\omega \\ ((C, \tilde{C}))_2 &= \int_{\Omega} K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} d\omega \\ \|U\| &= ((U, U))^{1/2} \\ (U, \tilde{U})_H &= \int_{\Omega} -v \tilde{v} - K_C C \tilde{C} d\omega \\ |U|_H &= |(U, U)_H|^{1/2} \end{aligned}$$

Step 2. Now the hydrostatic weak formulation of the polluted atmosphere can be brought to an analogous format as seen above:

$$\begin{aligned} & \int_0^T (U, \frac{d}{dt} \tilde{U})_H dt + \int_0^T a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U}) dt = \\ & = \int_0^T l(\tilde{U}) dt + \int_0^T \int_{\Omega} U_0 \tilde{U}(0) d\omega dt \end{aligned}$$

with

$$\begin{aligned}
a(U, \tilde{U}) &= a_1(U, \tilde{U}) + K_C a_2(U, \tilde{U}) \\
a_1(U, \tilde{U}) &= \int_{\Omega} \nabla_{\nu} v \nabla_{\nu} \tilde{v} \, d\omega \\
a_2(U, \tilde{U}) &= \int_{\Omega} K_2 \partial_2 C \partial_2 \tilde{C} + K_3 \partial_3 C \partial_3 \tilde{C} \, d\omega \\
b &= b_1 + K_2 b_2 \\
b_1(U, \tilde{U}, U^{\#}) &= - \int_{\Omega} (v \tilde{v} + v \tilde{w}) \nabla v^{\#} \\
b_2(U, \tilde{U}, U^{\#}) &= - \int_{\Omega} (v \tilde{C} + w \tilde{C}) \nabla C^{\#} \\
e(U, \tilde{U}) &= \int_{\Omega} b(v) \tilde{v} \, d\omega \\
l(\tilde{U}) &= - \int_{\Omega} S \tilde{C} \, d\omega.
\end{aligned}$$

Here the term $b(\mathbf{u}_H) = \alpha(-u_2, u_1)$ represents the cross product, and we use the fact that the vertical velocity w can be expressed as $w = w(v)$ because the divergence of the three dimensional velocity is zero.

Step 3. The next step is to check if the properties that yield the foundation of the proof still hold. It is easy to see that a is trilinear continuous, the coercivity of a is also trivial, moreover e is bilinear continuous and $e(U, U) = 0$.

What changes slightly is the proof of the results available on the expression b since the space derivatives in our case are applied on the test function.

- To prove the bound for $U, \tilde{U} \in V, U^{\#} \in V_{(2)}$, in this case we can not use the L_{∞} bound on the third term, since we would eventually arrive to $\|C^{\#}\|_{H^3}$, which means that we can not close the bound that way. We can use

$$\begin{aligned}
\int_{\Omega} w \tilde{C} \nabla C^{\#} \, d\omega &\leq c \|w\|_{L^2} \|\tilde{C}\|_{L^4} \|\nabla C^{\#}\|_{L^4} \leq \\
&\leq c \|U\| \|\tilde{U}\| \|\nabla C^{\#}\|_{H^1} \leq c \|U\| \|\tilde{U}\| \|U^{\#}\|_{V^{(2)}}
\end{aligned}$$

- For $U, U^\# \in V, \tilde{U} \in V_{(2)}$ we use

$$\int_{\Omega} w \tilde{C} \nabla C^\# d\omega \leq c \|\tilde{C}\|_{L^\infty} \|w\|_{L^2} \|\nabla C^\#\|_{L^2} \leq c \|U\| \|\tilde{U}\|_{V^{(2)}} \|U^\#\|,$$

exploiting the fact that in $n = 3$ we have $H^2 \subset L^\infty$.

- The proof of $b(U, \tilde{U}, U^\#) = -b(U, U^\#, \tilde{U})$ can be derived similarly as we derived

$$\int_0^T (\mathbf{u} \nabla C, \tilde{C}) dt = - \int_0^T (-\mathbf{u} C, \nabla \tilde{C}) dt,$$

using the integration by parts formula, observing that the boundary terms disappear, and finally using the fact that the divergence of the velocity is zero.

- The establishment of the improvement of the first bound works the same way as in the original article, except that we don't apply the switch provided by the result of the previous point, i.e. we perform the estimate

$$\int_{\Omega} w \tilde{C} \nabla C^\# d\omega \leq \|w\|_{L^2} \|\tilde{C}\|_{L^3} \|\nabla C^\#\|_{L^6},$$

and after this everything follows the same way.

This means that all the basic results and properties of the functions a, b, e still hold in this version.

Step 4. In this step we provide the operator form of the equation.

The change in this part of the proof is that we will need an additional operator Z for the time derivative part. In the original version $\frac{dU}{dt}$ was used as an operator acting on $\tilde{U}$, the result defined to be their inner product. In this version, because we apply the weak formulation in space-time, the time derivation is applied on the test function, so we will need the following operator Z :

$$\langle Z(U), \tilde{U} \rangle = \int_0^T (U, \frac{d\tilde{U}}{dt})_H dt.$$

With this the operator form of the equation is

$$ZU + AU + B(U, U) + EU = l + U_0,$$

where all the other operators are understood in a way that the time integration is included. Since in this version we work with both time-space and only-space (stationary case) weak formulation as well, it is always assumed that we use the one according to the context.

Step 5. The stationary case works as it worked in the original article, since the operator form of the new version is created in a way that ZU and U_0 contains all the time-related part, after dropping it we arrive to the same equation as in the original case.

Step 6. Finally we discuss the details of the time-dependent case. Once again we remind ourselves that in this case, since in our version we performed the weak formulation in time as well, we can not use notations such as $\frac{dU_k}{dt}$. Instead, we will use ZU_k , which as an operator acting on $\tilde{U}$ yields the inner product of the function and the time derivative of the test function.

Even though the time derivative here is applied on the test function, using the time difference for the solution function does work, in fact the original article arrives from the same start to a weak formulation where the derivative is applied on the test function.

Basically with the time discretization form of

$$\frac{1}{\Delta t}(U^n - U^{n-1}, \tilde{U})_H + a(U^n, U^n) + b(U^n, U^n, \tilde{U}) + e(U^n, \tilde{U}) = l(\tilde{U}),$$

we arrive to the expression

$$\begin{aligned} - \int_0^T (U, \tilde{U})_H \psi' dt + \int_0^T [a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U})] \psi dt = \\ = (U_0, \tilde{U}) \psi(0) + \int_0^T l(\tilde{U}) \psi dt, \end{aligned}$$

where ψ is a function defined on $[0, T]$, used to create the time-dependent version $\tilde{U}\psi$ of the test function. If we rename this product simply to be $\tilde{U}$, the time-dependent test function, we arrive exactly to,

$$\begin{aligned} - \int_0^T (U, \partial_t \tilde{U})_H dt + \int_0^T [a(U, \tilde{U}) + b(U, U, \tilde{U}) + e(U, \tilde{U})] dt = \\ = (U_0, \tilde{U}(0)) + \int_0^T l(\tilde{U}) dt, \end{aligned}$$

which is what we needed.

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